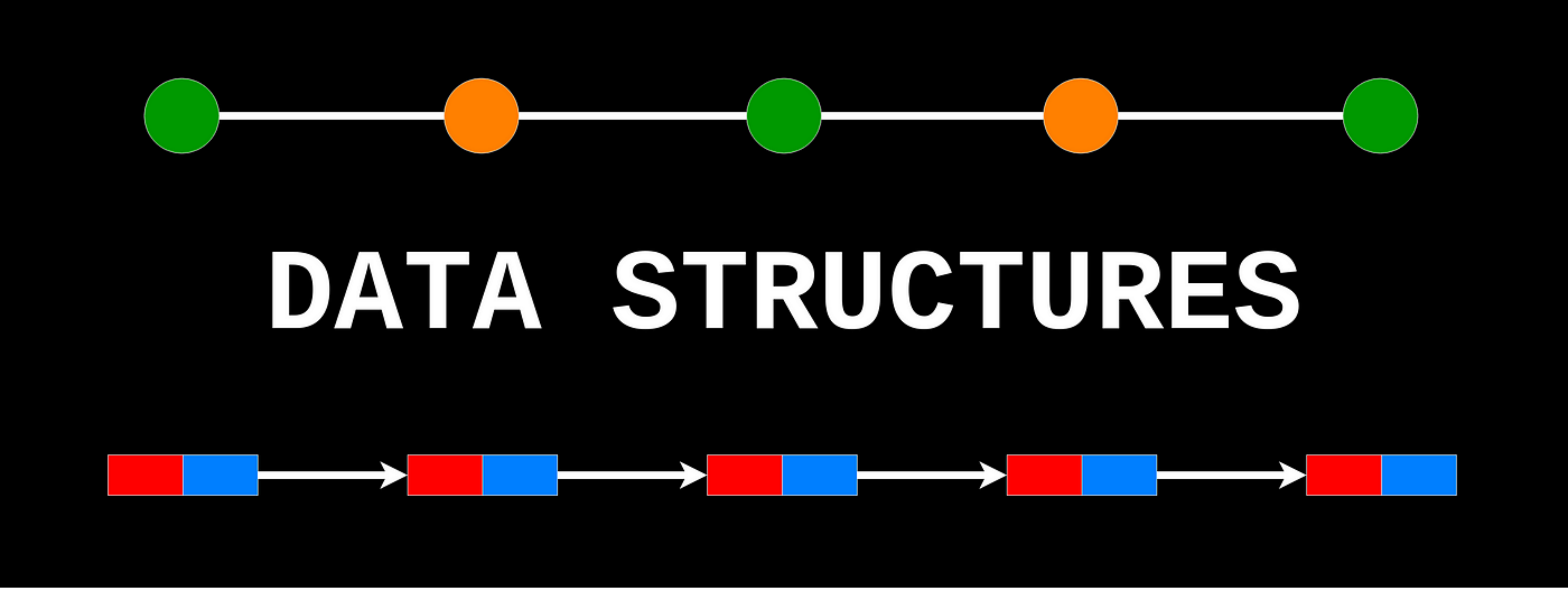
Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

OOP in Python – Data Structures

Welcome to *the* Object Oriented Programming *course*

1 - Data Structures



- A **data structure** is a **storage** that is used to **store** and **organize data**. It is a way of arranging data on a computer so that it can be accessed and updated efficiently.

- **What is a data structure?** Dalam istilah ilmu komputer, sebuah struktur data (*data structure*) adalah cara **penyimpanan**, **penyusunan** dan **pengaturan data** di dalam media penyimpanan komputer sehingga data tersebut dapat digunakan secara efisien.

- A **<u>data structure</u>** is not only used for organizing the data. It is also used for **processing**, **retrieving**, and **storing data**.

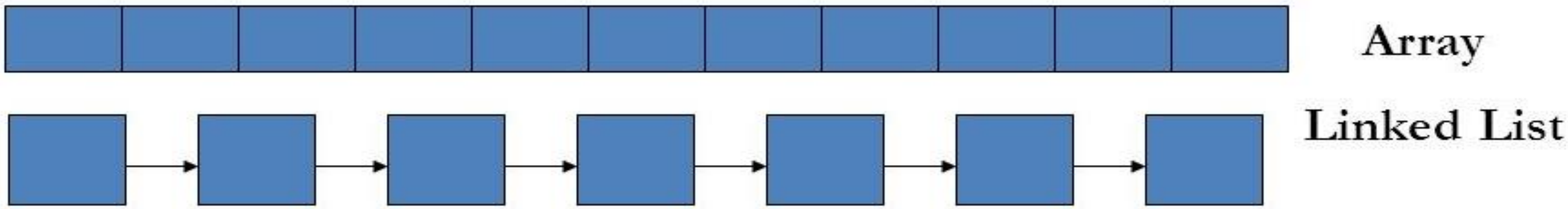
- There are different basic and advanced types of data structures that are used in almost every program or software system that has been developed.

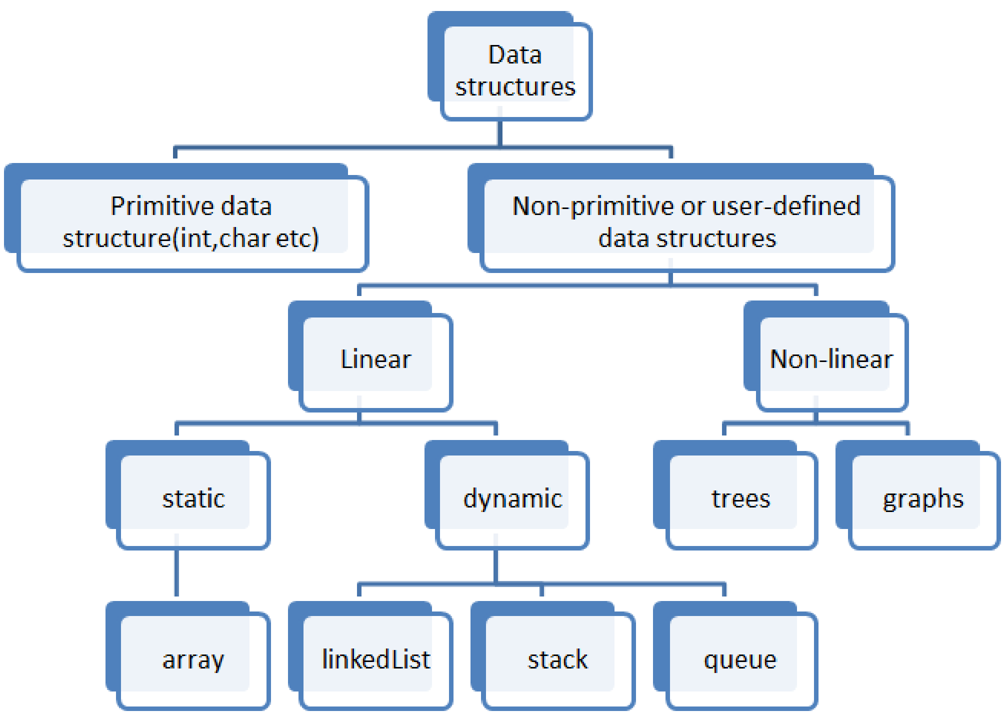
Popular types of Data Structures:

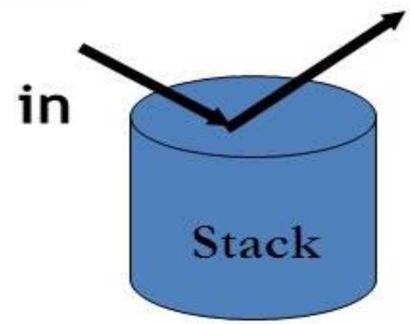
- Array
- Linked List
- Stack
- Queue
- Binary Tree
- Binary Search Tree
- Heap
- Hashing
- Graph
- Matrix
- Misc
- Advanced Data Structure

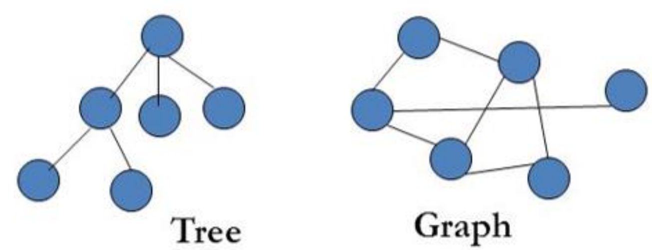
What *is* Data Structure?

Data may be organized in different ways. We can represent the data in a particular way, so that it can be used efficiently.









What *is* a Data Structure?

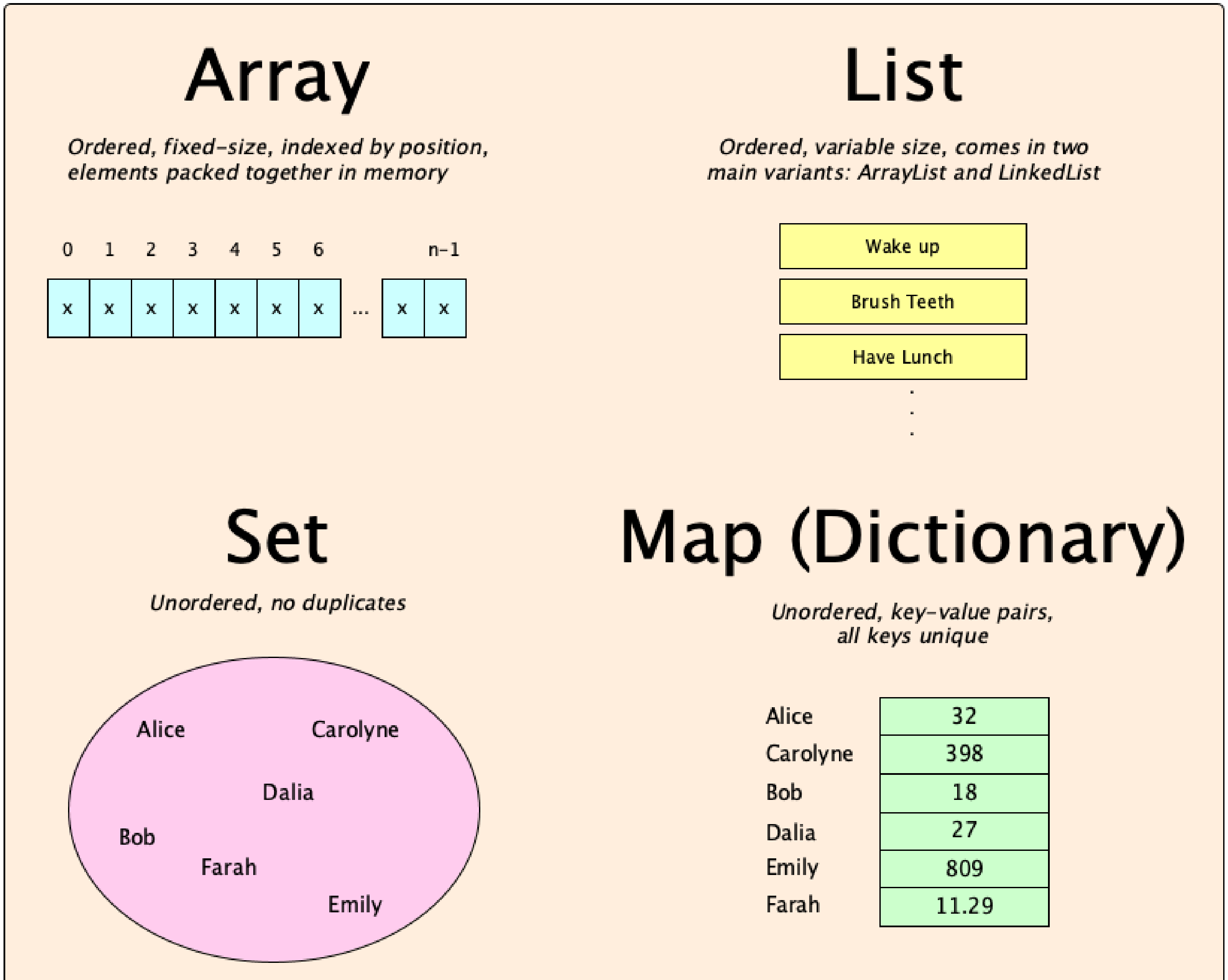
❖ **<u>Data structure</u>** is a collection of data that has specific ways of **accessing**, **storing**, and **organizing** the data. A data structure also defines any relationship between multiple pieces of data. It's like a **container**.

❖ A data structure (**container**) that has a ordered series of numbered slots that can each contain a piece of data. We access a piece of data with an index. Usually stores homogeneous data.

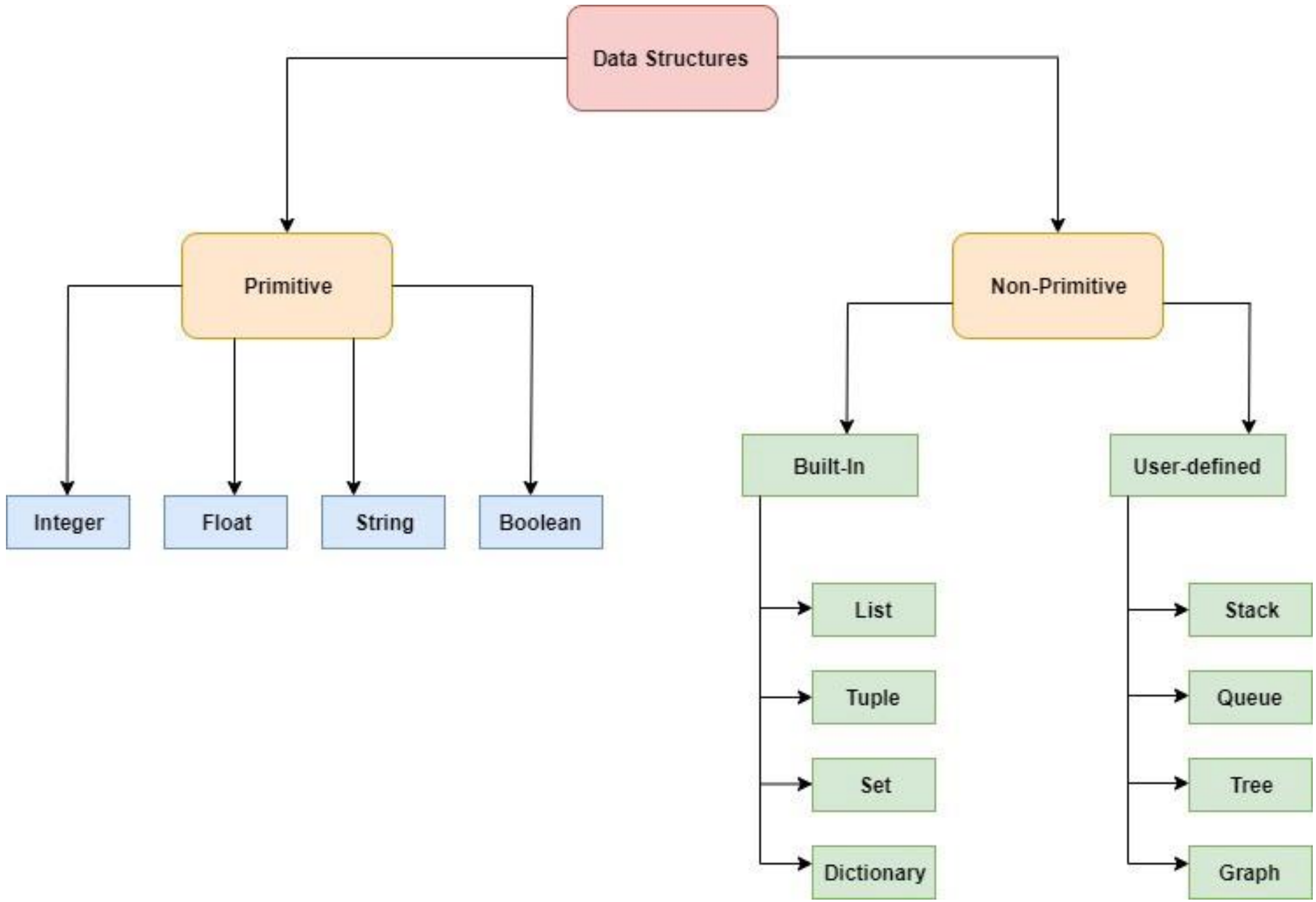
40	55	63	17	22	68	89	97	89	
0	1	2	3	4	5	6	7	8	<- **Array Indices**

Array Length = 9
First Index = 0
Last Index = 8

Java Data Structures



Python Data Structures



Data structure is the way data is arranged, stored, and organized in a computer program.

Many security algorithms today, seperti <u>algoritma enkripsi</u>, <u>hashing</u>, atau <u>autentikasi</u>, melibatkan penggunaan <u>**struktur data khusus**</u> untuk mengelola dan memproses data dengan cara yang aman dan efisien.

Data structure is used to represent and store data used in training and testing models. Efficient data representation can affect the **performance and speed** of **machine learning algorithms**.

An efficient data structure enables **fast access and manipulation of data**, which is essential in processing **large datasets**. Machine learning and deep learning algorithms require fast access to optimize model training and inference time.

AlphaGo is an AI program developed by DeepMind (March 2016), using data structures as an integral part of its design and implementation. AlphaGo uses multiple neural networks to make decisions and predictions in the game of Go. Data structures are used to store the weights and architecture of these neural networks.

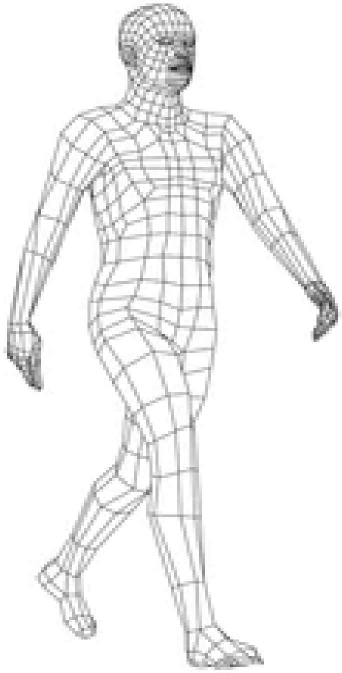
ChatGPT developed by OpenAI and announced by OpenAI in June 2020. GPT-3.5 which is the basis for ChatGPT is a model based on the GPT-3 (Generative Pre-trained Transformer version 3) architecture. ChatGPT uses various data structures as part of its implementation. The data structures are used to store and manage the information required in the context of the conversation, to keep track of the context, understand the questions, and generate appropriate responses.

Orang-orang hebat seperti Elon Musk, Mark Zuckerberg, Sam Altman, dan Sundar Pichai tidak hanya terampil dalam pemahaman konsep dasar seperti *data structure*, tetapi mereka juga memiliki tim ahli dalam pengembangan *smart system*. Penggunaan *data structure* dalam pengembangan *advanced* dan *smart system* seperti yang mereka lakukan dapat mencakup "**algoritma dan optimisasi**", **pengolahan big data**, **manajemen memori**, "**optimisasi database**", "*artificial intelligence and machine learning*" dan bahkan pada "**pengembangan framework dan library**".

2 - What *is* data structure in OOP?

Data Structure *in* OOP

Object Oriented Programming (OOP) adalah tentang <u>memperlakukan data</u> sebagai objek (*objects*). Segala sesuatu dalam hidup dapat direpresentasikan sebagai objek dalam *code*. Namun, kita harus memiliki **blueprint (Class OOP)** sebelum kita membangun apapun.



Blueprint Person

Blueprint (called a class in code): **Person**

1) String Name
2) Integer Age
3) String Eye_Color

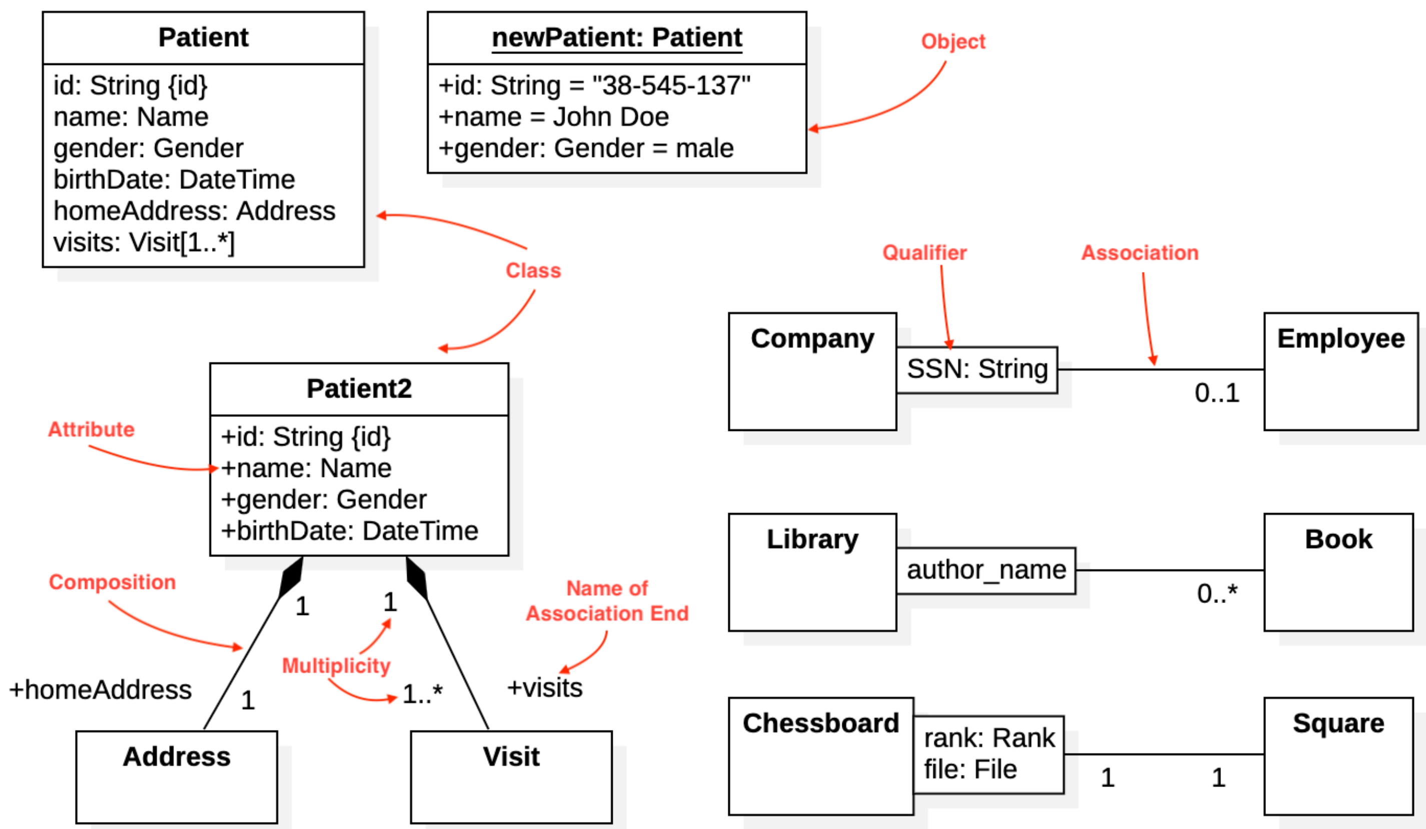
- The **<u>blueprint contains the structure that will hold the pieces of data</u>**, we care about for a given person. We can create a person with something called a **constructor**, with a special type of function that with allow us to **create people** (*also called people instances*).

- A **blueprint** or **class** contains **properties** (name, age, eye color), a **constructor**, and **behavior** (*functions that access/alter the values of our properties or calculate related things*).

- These **<u>behaviors</u>** <u>or</u> **<u>functions</u>** are often called **methods** because they are tied/involved to the class/blueprint.

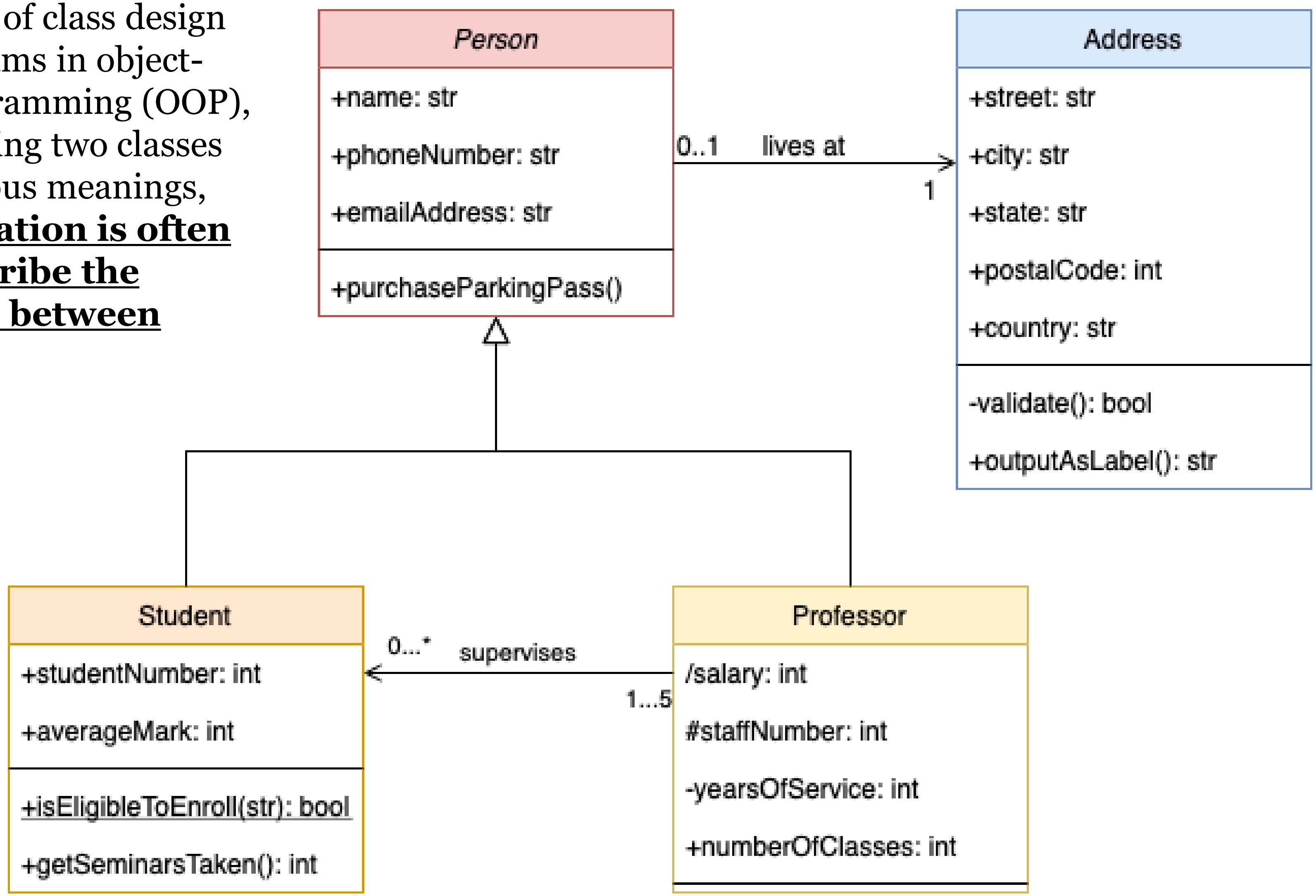
Data Structure *in* OOP

- The **blueprint contains the structure that will hold the pieces of data**. A blueprint or class contains properties (*name, age, eye color*), a constructor, and behavior/function.
- **Object-Oriented Programming (OOP)** and **class diagrams** in the **Unified Modeling Language (UML)** are interrelated. UML is a **visual modeling language** used to **document**, **design**, and **model** software systems, including systems built with the OOP paradigm.
- One type of UML diagram used to show the static structure of a software system is the **UML class diagram**. In this context, classes in OOP are very important.

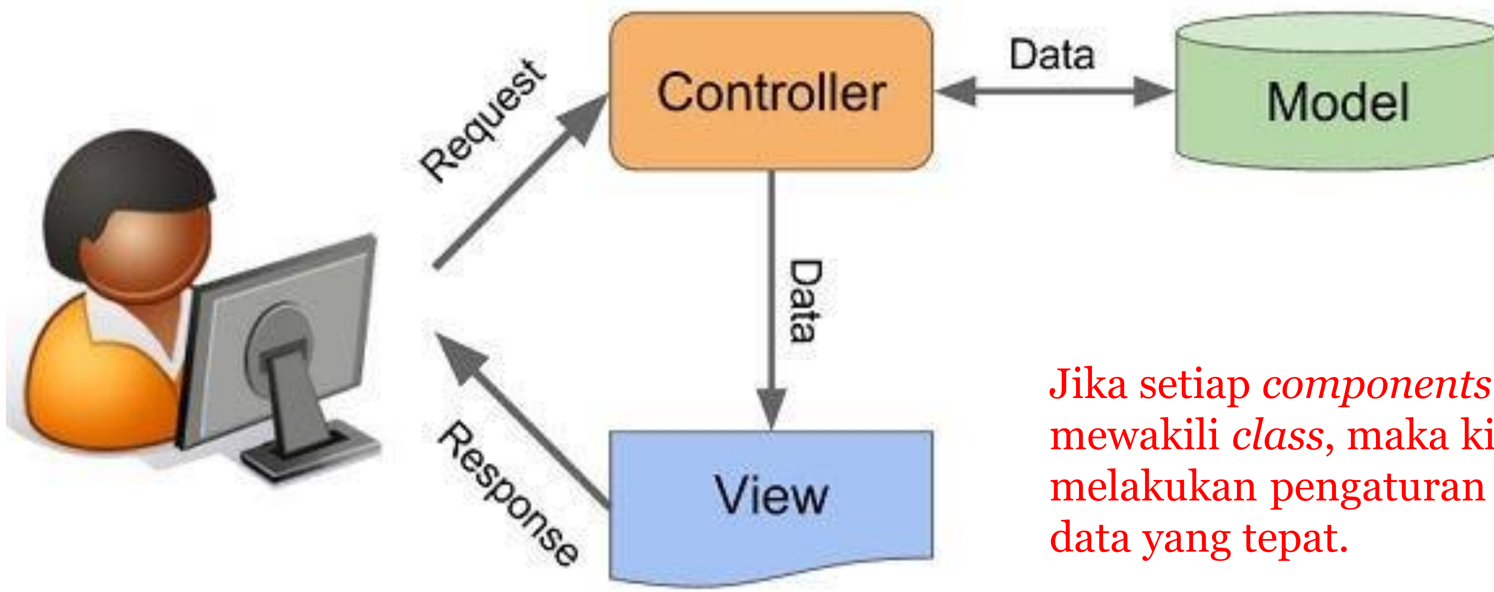


Example Class Diagram

In the context of class design or class diagrams in object-oriented programming (OOP), a line connecting two classes can have various meanings, **<u>and this notation is often used to describe the relationship between classes</u>**.



Perlu untuk diingat pada bagian awal ini, yaitu tujuan utama *data structure* adalah untuk memungkinkan pengaksesan dan manipulasi data dengan efisien.



Jika setiap *components* akan mewakili *class*, maka kita harus melakukan pengaturan pengaksesan data yang tepat.

- Dalam OOP, sebuah *object* akan memiliki **atribut (data)** dan **method/metode (fungsi)** yang bekerja pada data.

- Paradigma pemrograman OOP ini menggunakan konsep *object*, dimana menggabungkan **DATA** dan **FUNGSI** terkait ke dalam satu entitas yang disebut *object*.

- Karena *object* dalam OOP sering kali menggunakan *data structure* untuk menyimpan dan mengelola data internal. Contohnya, *object* dapat memiliki *attribute* yang mewakili data, dan *method* yang melakukan operasi pada data tersebut.

Test your understanding

The following are some multiple choice questions to reinforce your understanding of the concepts we've covered so far.






Question #1

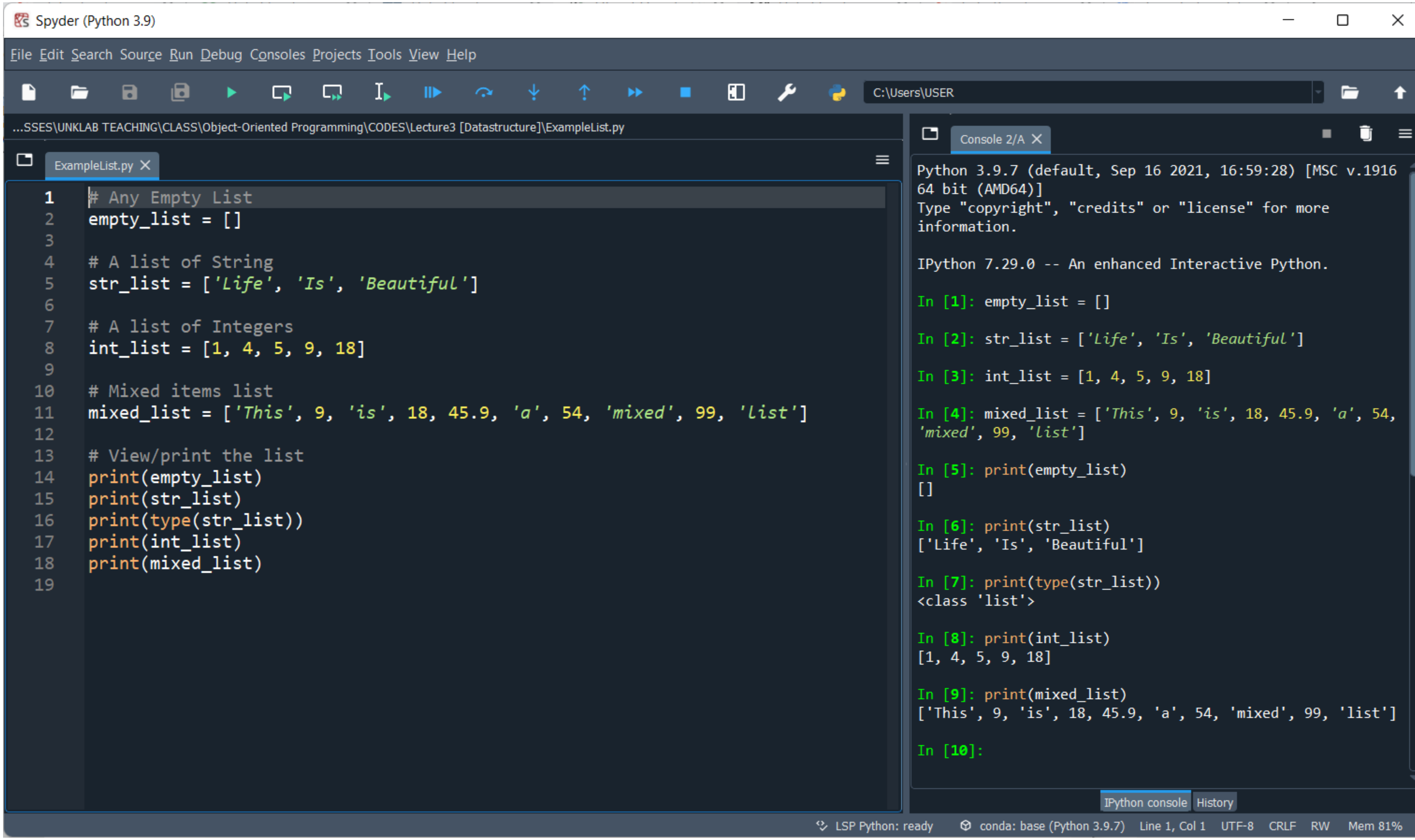
Apa yang di maksud dengan **data structure** atau struktur data? Jelaskan hubungan data structure dan object oriented programming (OOP)?

Question #2

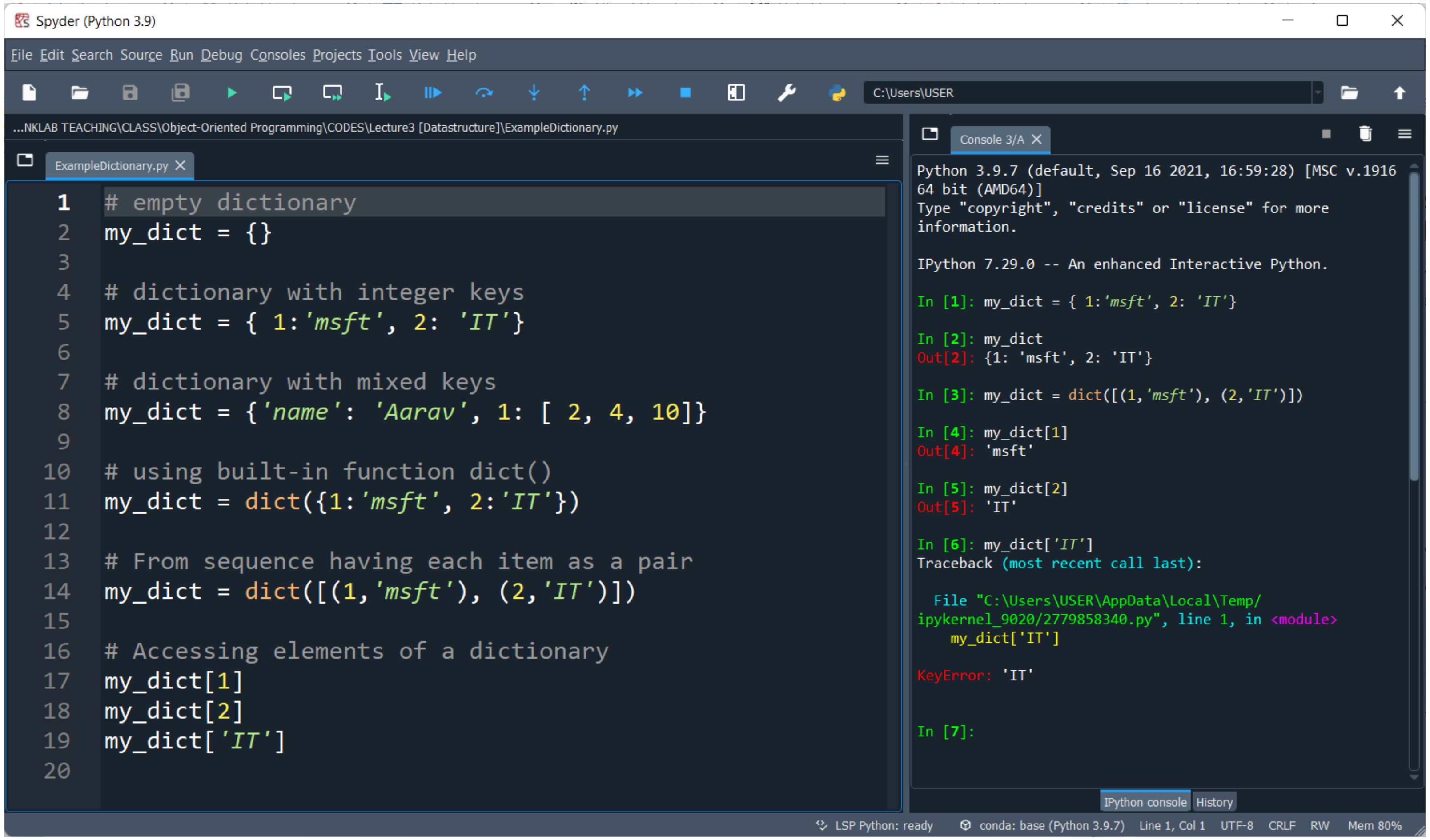
Object Oriented Programming (OOP) adalah tentang memperlakukan data sebagai objek (*objects*). Blueprint atau disebut juga Class (*di dalam code program*) akan berisi **properties** (name, age, eye color), **constructor** dan juga **behaviors (functions atau methods)**. Sebutkan contoh **behaviors** dari class Person?

Exercise *for* Students

[1] Data Structure - List



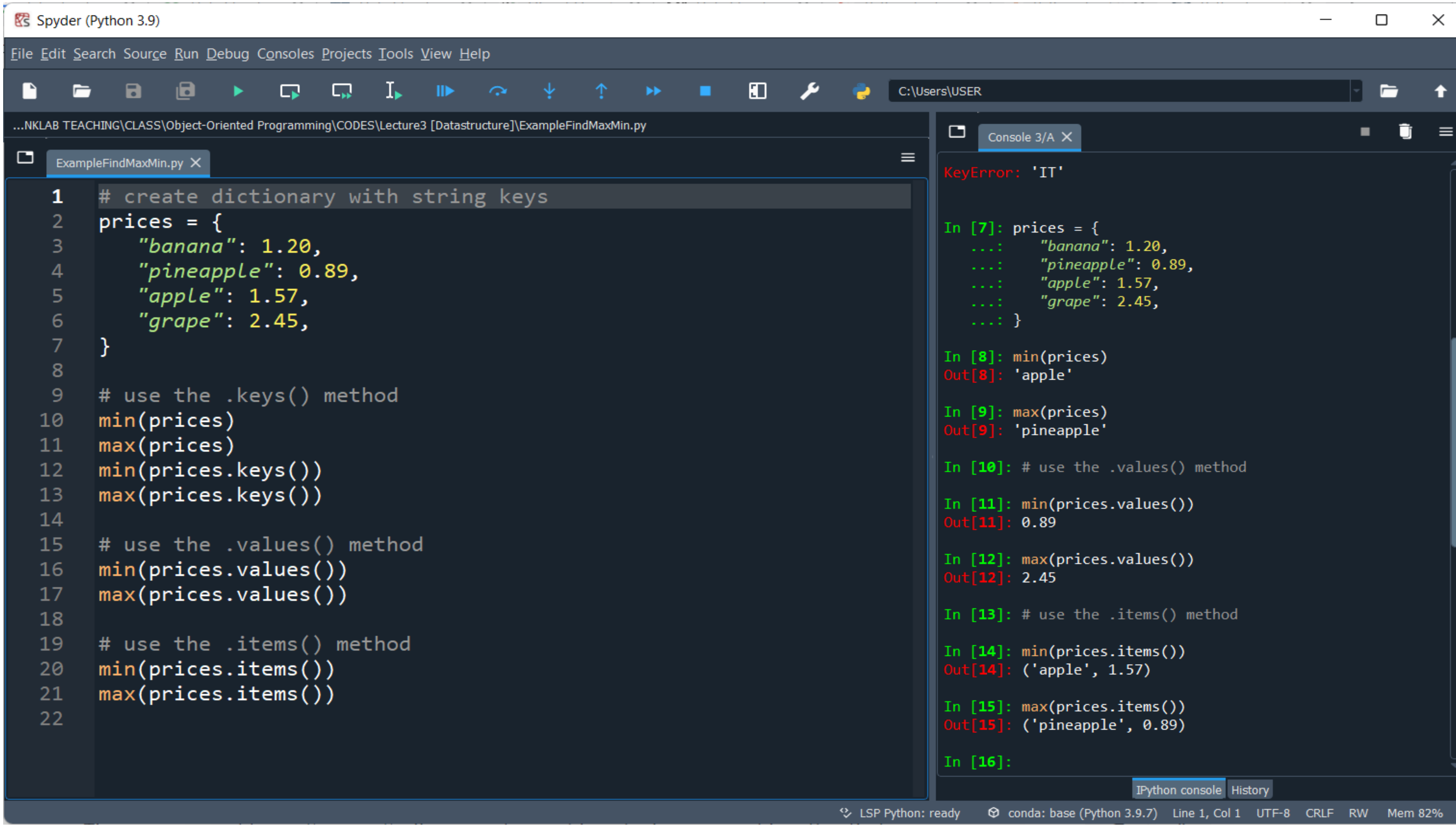
[2] Data Structure - Dictionary



Jelaskan mengapa terjadi Error???

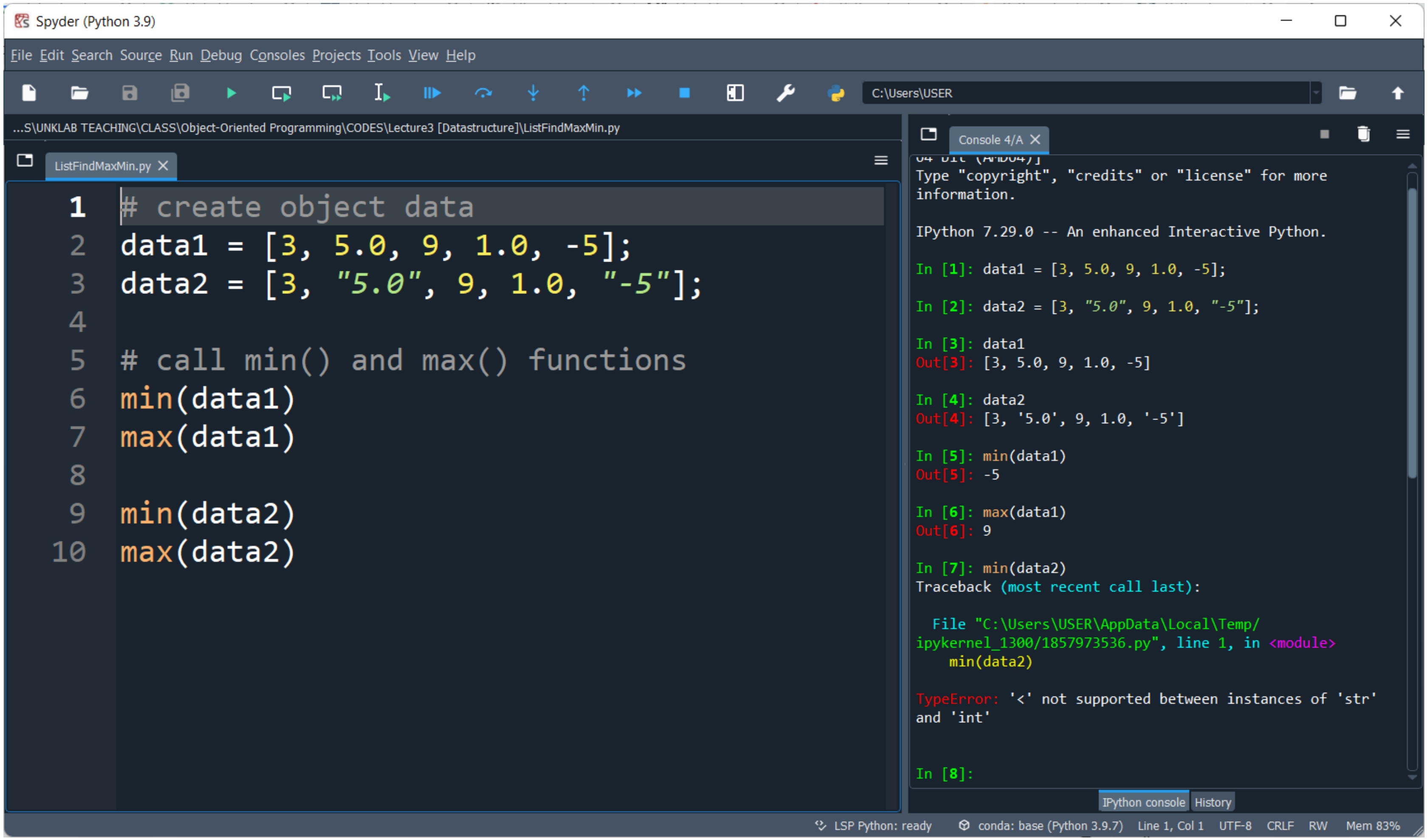
[3] Processing Dictionaries With min() & max()

When it comes to processing Python dictionaries with min() and max(), we need to consider that if we use the dictionary directly, then both functions will operate on the keys:



[4] Find smallest & largest values *in* List

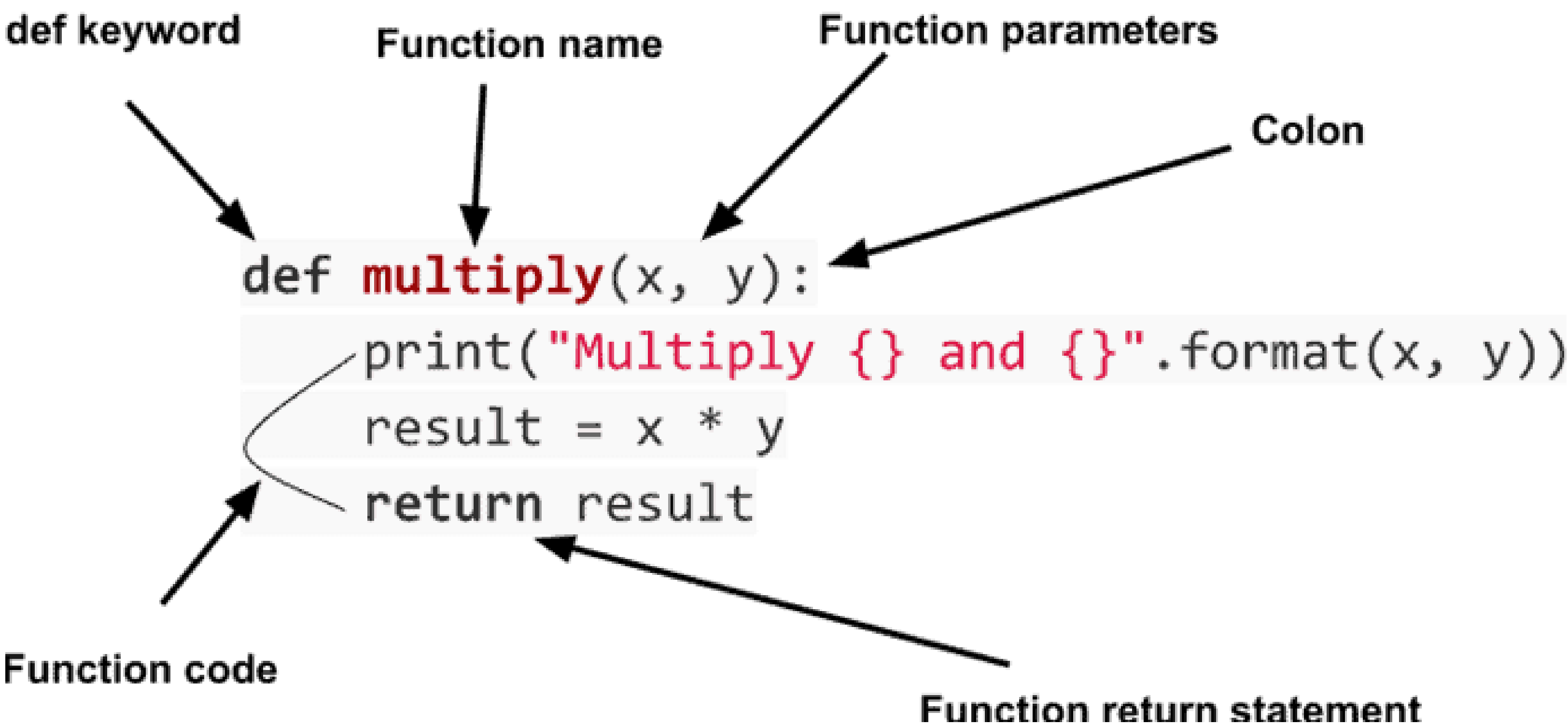
Use Python’s min() and max() to find smallest and largest values in List data.



Jelaskan mengapa terjadi Error???

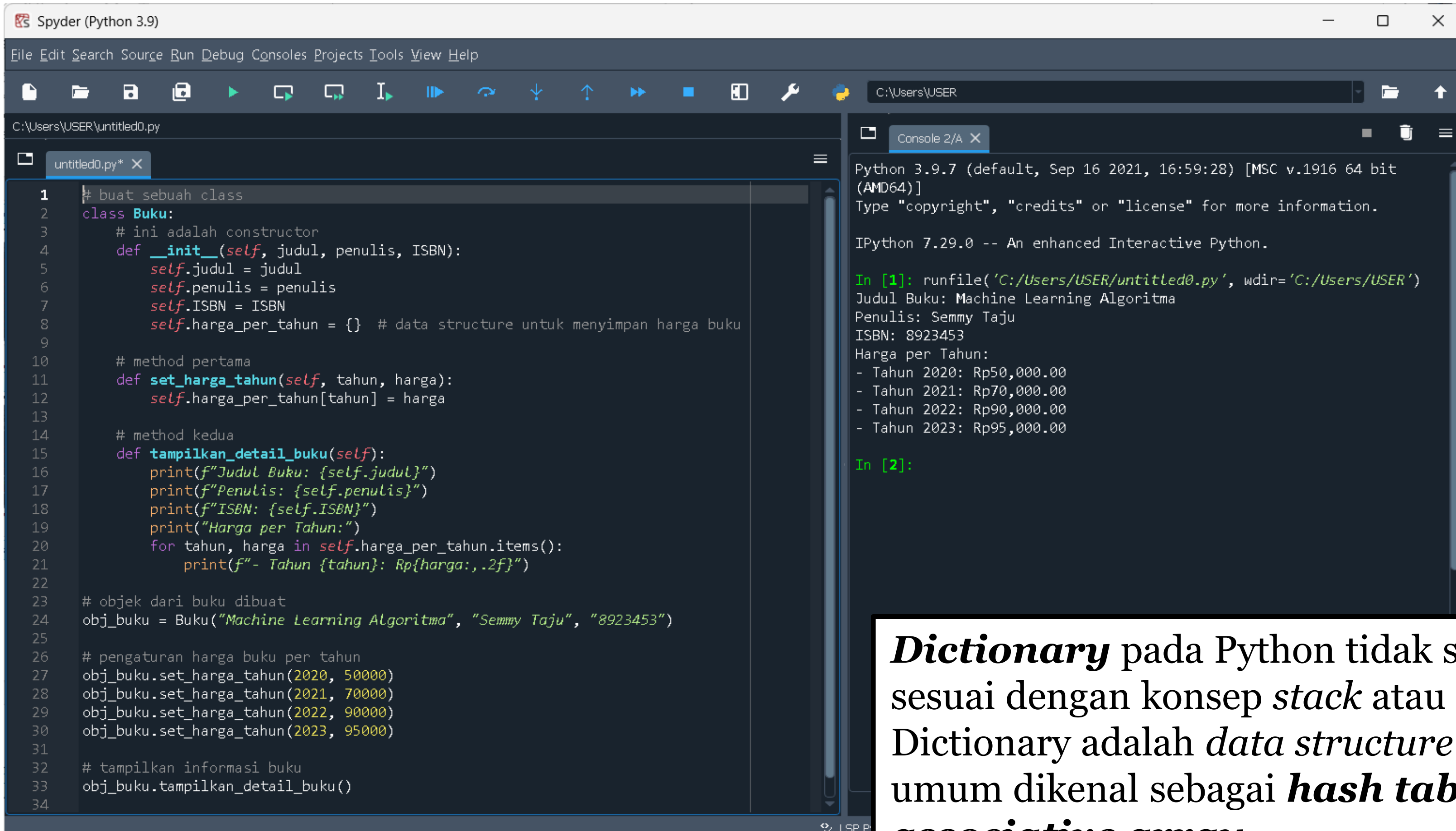
Python Functions

A **Python function** is a block of code which only runs when it is called. You can pass data, known as parameters, into a function. **“*def*” keyword** pada Python adalah singkatan dari "*define*" yang **<u>menunjukkan bahwa kita sedang mendefinisikan</u>** sebuah fungsi.



[5] Data Structure *in* Class

The *data structure* used is dictionary (*harga_per_tahun*). Dictionary is a *data structure* that allows us to store key-value pairs.



Dictionary pada Python tidak sepenuhnya sesuai dengan konsep *stack* atau *queue*. Dictionary adalah *data structure* yang lebih umum dikenal sebagai ***hash table*** atau ***associative array***.

Perhatikan class OOP di atas, jelaskan *structure* dari class OOP tersebut?

END PRESENTATION

Thank you for your attention
Instructor: S – W – T